

A 1/4 Incidence Identity on the 600-Cell and its Bekenstein–Hawking Analogy

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May 2026

Status: Preprint (not peer-reviewed).

Abstract

We prove a finite-arithmetic identity on the binary icosahedral group $2I$ (equivalently, the 120 vertices of the 600-cell V_{600}). Let $\text{Dic}_5 \subset 2I$ denote the binary dihedral subgroup of order 20 generated by the bulk T_τ -cycles, and let $V_{24} \subset V_{600}$ denote the 24-cell sublattice fixed by the Galois twist $\sigma : \sqrt{5} \mapsto -\sqrt{5}$ acting on the icosian quaternion components. We show that $|gH \cap V_{24}| = 4$ and $|Hg \cap V_{24}| = 4$ for every left coset gH and right coset Hg of $H = \text{Dic}_5$ in $2I$, where Dic_5 is self-normalizing (and hence non-normal) in $2I$. Hence each coset partitions into 4 σ -fixed and 16 σ -mobile vertices, giving the rational identity

$$\frac{S(C)}{A(C)} = \frac{4}{16} = \frac{1}{4}$$

for every coset C , where $S(C) := |C \cap V_{24}|$ and $A(C) := |C \setminus V_{24}|$. The identity is moreover τ -independent: every choice of order-10 element $\tau \in 2I$ yields a bulk subgroup $H_\tau \cong \text{Dic}_5$, and all such choices give the same per-coset incidence. Interpreting S as a count of twin-anchored information bits and A as a count of dynamical horizon channels, the structural ratio has the same dimensionless numerical value as the Bekenstein–Hawking entropy coefficient. The result is parameter-free, established by exact $\mathbb{Q}(\sqrt{5})$ enumeration, and is independent of any cosmological, dynamical, or quantum-gravity hypothesis. We make no claim regarding cascade-unit calibration to physical Planck units; the rational ratio is the load-bearing content.

1 Introduction

The Bekenstein–Hawking formula, in natural units $\hbar = G = c = k_B = 1$, $S_{\text{BH}} = A/4$, [Bekenstein, 1973, Hawking, 1975] attaches a thermodynamic entropy to a black-hole horizon proportional to its area, with the dimensionless constant of proportionality $1/4$. The coefficient $1/4$ is universal across asymptotic black-hole spacetimes, and its microscopic origin has been a central question of quantum gravity.

Two prominent microscopic derivations have been proposed: brane state-counting in string theory [Strominger and Vafa, 1996] and isolated-horizon spin-network counting in loop quantum gravity [Ashtekar et al., 2000]. Each yields the $1/4$ coefficient (modulo standard normalisation conventions in the LQG case) via a partition function or character formula that involves non-trivial dynamical machinery.

This paper reports a finite-arithmetic identity that produces the same dimensionless numerical value by counting alone. Specifically, the count $|gH \cap V_{24}|$ equals 4 for every coset of a particular order-20 subgroup H of the binary icosahedral group, and the complementary count $|gH \setminus V_{24}|$ equals 16. The rational ratio $4/16 = 1/4$ is therefore an exact structural identity

on a finite group. We do *not* claim a derivation of physical black-hole thermodynamics; we report a structural ratio that matches the universal coefficient under a transparent interpretive identification of the two coset components.

The remainder of the paper is organised as follows. Section 2 fixes notation and the exact arithmetic in $\mathbb{Q}(\sqrt{5})$ used throughout. Section 3 identifies the bulk subgroup $\text{Dic}_5 \subset 2I$ and gives an explicit binary-dihedral presentation certificate. Section 4 verifies the σ -fixed sublattice V_{24} . Section 5 states and proves the main coset-incidence theorem. Section 6 reports the $1/4$ structural coefficient. Sections 7 and 8 mark the scope and place the result alongside existing microscopic derivations.

2 Exact model

All arithmetic in this paper is exact. Coordinates of V_{600} vertices lie in the field $K := \mathbb{Q}(\sqrt{5})$, represented as ordered pairs $(a, b) \in \mathbb{Q}^2$ standing for $a + b\sqrt{5}$. The Galois automorphism $\sigma : K \rightarrow K$ acts by $\sigma(a + b\sqrt{5}) = a - b\sqrt{5}$.

Definition 2.1 (Icosian quaternions and V_{600}). *Let $\mathbb{H}_K$ be the Hamilton quaternion algebra over K and $\varphi = (1 + \sqrt{5})/2$. The 600-cell V_{600} is the set of 120 unit icosians obtained as the union of the following three coordinate orbits with respect to the standard $\mathbb{H}_K$ basis:*

- (i) *the 8 axis vertices $(\pm 1, 0, 0, 0)$ and the 3 permutations thereof;*
- (ii) *the 16 half-type vertices $(\pm 1/2, \pm 1/2, \pm 1/2, \pm 1/2)$;*
- (iii) *the 96 even-permutation vertices: vectors whose four coordinates are an even permutation of $(0, \pm 1/(2\varphi), \pm 1/2, \pm \varphi/2)$ with at least one zero, and the three nonzero coordinates ranging over all $2^3 = 8$ sign combinations.*

This is the standard construction of the binary icosahedral group $2I \subset \text{Sp}(1, K)$ [Conway and Sloane, 1988, Coxeter, 1973, Moody and Patera, 1993].

The map σ extends componentwise to icosian quaternions; it sends V_{600} bijectively to a second 600-cell $\sigma(V_{600})$ that meets V_{600} exactly along the 24 axis- and half-type vertices identified in Section 4.

Lemma 2.1 (120 vertices, group closure, unit norm). *The set V_{600} has exactly 120 elements; every element has K -norm 1; and V_{600} is closed under quaternionic multiplication, forming the binary icosahedral group $2I$.*

Proof. Direct enumeration of axis (8), half-type (16), and even-permutation (96) coordinate orbits gives 120 exact icosian vectors in $\mathbb{Q}(\sqrt{5})^4$; deduplication confirms cardinality. Unit norm $|v|^2 = 1$ and exhaustive group closure across all 14,400 products $v_i v_j$ are checked by the accompanying `verify.py` certificate (cf. Conway and Sloane, 1988). $\square$

We use $G := 2I = V_{600}$ and $|G| = 120$ throughout.

3 The bulk subgroup

Definition 3.1 (T_τ -cycles). *For $\tau \in G$ with $\text{ord}(\tau) = 10$ (there are exactly 24 such elements; pick any one), left multiplication by τ generates a permutation $T_\tau : G \rightarrow G$, $g \mapsto \tau g$, whose cycle decomposition partitions G into 12 cycles each of length 10. We will later show (Theorem 5.2) that the eventual incidence identity is independent of the choice of τ .*

Definition 3.2 (K-classes). For each vertex $v \in G$, write $\text{shell}(v)$ for the index (0 through 8) of the rational squared-distance class $|v - 1|^2 \in K$ in the totally ordered set of distinct shell radii; ordering is by exact sign comparison on K (no floating point). For a cycle C , set $K(C) := \sum_{v \in C} (\text{shell}(v) - 4)^2$.

Lemma 3.1 (K-multiset). The 12 cycles distribute into the K -multiset $\{72:1, 0:1, 52:5, 20:5\}$: one cycle with $K = 72$ (the maximum), one with $K = 0$ (vertices uniformly on shell 4), five with $K = 52$, and five with $K = 20$.

Proof. Direct rational computation on G . □

Definition 3.3 (Bulk subgroup H). Let $H \subset G$ be the union of the two K -special cycles ($K = 72$ and $K = 0$). Then $|H| = 20$.

Lemma 3.2 ($H = \text{Dic}_5$). H is closed under quaternionic multiplication, contains the identity, and admits the binary-dihedral presentation $H = \langle a, b \mid a^{10} = 1, b^2 = a^5, bab^{-1} = a^{-1} \rangle \cong \text{Dic}_5$.

Proof. Closure and presence of $1 \in H$ are verified by exhaustive enumeration across all 400 products in $H \times H$. The generator search returns an explicit pair $(a, b) \in H \times H$: a is an order-10 element drawn from the $K = 72$ cycle, and b is one of ten elements of H satisfying $b^2 = a^5$ and $bab^{-1} = a^{-1}$ (we fix one such choice). The accompanying certificate then verifies $\langle a, b \rangle = H$ by enumerating the closure of $\{a, b\}$ under quaternionic multiplication and comparing it to H . □

Lemma 3.3 (Coset cycle composition). For $T_\tau : g \mapsto \tau g$ with $\tau \in H$, every non-bulk right coset Hg is the disjoint union of one whole T_τ -cycle of K -class 52 and one whole T_τ -cycle of K -class 20. The five non-bulk right cosets therefore exhaust the ten non-bulk T_τ -cycles. By contrast, every non-bulk left coset gH contains exactly two vertices from each of the ten non-bulk cycles, hence 10 vertices of K -class 52 and 10 of K -class 20 in aggregate but is not a union of whole T_τ -cycles.

Proof. Right cosets: $T_\tau(Hg) = (\tau H)g = Hg$ because $\tau \in H$ implies $\tau H = H$ (closure of H). Therefore Hg is closed under T_τ and decomposes as a disjoint union of whole T_τ -cycles; the cardinality count $|Hg| = 20 = 10 + 10$ together with the K -class assignment gives one cycle of each non-bulk K -class.

Left cosets: $T_\tau(gH) = \tau gH$, which equals gH iff $g^{-1}\tau g \in H$. The certificate `verify.py` checks the finite condition $g^{-1}\tau g \notin H$ for every $g \in G \setminus H$ explicitly. This direct enumeration (not derivable from self-normality of H alone) establishes that T_τ does not preserve any non-bulk left coset. The induced action of T_τ on the five non-bulk left cosets is therefore a permutation with no fixed points; since $\tau^{10} = 1$ acts trivially on cosets and the order divides 10, the induced permutation has order 5, i.e. it is a 5-cycle. Hence the T_τ -orbit of any vertex $v \in gH$ has length 10 and visits each of the five non-bulk left cosets exactly twice, giving the 2-per-cycle distribution. The aggregate K -class composition 10+10 in each left coset follows immediately.

The whole-cycle / 2-per-cycle distinction is verified by the shared tests `test_non_bulk_left_cosets_2_per` and `test_non_bulk_right_cosets_pair_K52_K20`. □

Lemma 3.4 (H is self-normalizing). For every $g \in G \setminus H$, the left coset gH and the right coset Hg are distinct subsets of G . Hence $N_G(H) = H$, and H is self-normalizing and not normal in G .

Proof. Direct enumeration: there are $|G| - |H| = 100$ elements outside H , and the certificate `verify.py` confirms $gH \neq Hg$ for each of them. Therefore the only g with $gH = Hg$ are those in H itself, giving $N_G(H) = H$. Self-normalizing subgroups are non-normal unless they coincide with G , which H does not. □

A consequence of Lemma 3.4 is that the main incidence result of Section 5 cannot be inferred from a quotient construction: it must be checked on each coset family separately. We do so explicitly.

4 The σ -fixed 24-cell

Definition 4.1 (Galois twist on icosians). σ acts componentwise on icosian quaternions: for $v = (v_w, v_x, v_y, v_z)$ with each $v_i \in K$, $\sigma(v) := (\sigma(v_w), \sigma(v_x), \sigma(v_y), \sigma(v_z))$.

Definition 4.2 (24-cell as σ -fixed sublattice). $V_{24} := \{v \in V_{600} : \sigma(v) = v\}$.

A vertex $v \in V_{600}$ lies in V_{24} if and only if every coordinate has zero $\sqrt{5}$ -component, i.e. if every coordinate is rational.

Lemma 4.1 ($|V_{24}| = 24$). V_{24} has cardinality 24 and consists of the 8 axis points $(\pm 1, 0, 0, 0)$ (and permutations) and the 16 half-type points $(\pm 1/2, \pm 1/2, \pm 1/2, \pm 1/2)$. These 24 points form the exceptional regular 24-cell, the only regular convex 4-polytope without a 3-dimensional analogue [Coxeter, 1973].

Proof. The axis points are coordinatewise rational. The half-type points have coordinates $\pm 1/2$. The 96 even-permutation points have at least one coordinate involving $\varphi/2 = (1 + \sqrt{5})/4$ or $1/(2\varphi) = (\sqrt{5} - 1)/4$, hence a non-zero $\sqrt{5}$ component. Direct enumeration confirms $8 + 16 = 24$. $\square$

5 The incidence theorem

Theorem 5.1 (Coset incidence). Let $G = 2I$ and $H = \text{Dic}_5 \subset G$ as in Lemma 3.2, and let $V_{24} \subset G$ be the σ -fixed 24-cell of Lemma 4.1. Then for every left coset gH and every right coset Hg of H in G ,

$$|gH \cap V_{24}| = |Hg \cap V_{24}| = 4,$$

and consequently

$$|gH \setminus V_{24}| = |Hg \setminus V_{24}| = 16.$$

Proof. $|H| = 20$ (Lemma 3.2), so each coset has cardinality 20 by Lagrange. There are $|G|/|H| = 6$ left cosets and 6 right cosets, partitioning G into $6 \times 20 = 120$ vertices.

The intersection $H \cap V_{24}$ has exactly 4 elements (direct enumeration: the identity 1, the central element of order 2, and two elements of order 4, all coordinatewise rational). Let C be a non-bulk left coset; we show $|C \cap V_{24}| = 4$ by exact $\mathbb{Q}(\sqrt{5})$ enumeration of C and explicit intersection with V_{24} . The verification script `verify.py` performs this enumeration over both coset families, exhibits the explicit four-element intersection by vertex index in each case, and checks that the six left cosets (resp. six right cosets) partition G . The result is summarised in Table 1.

Because H is non-normal (Lemma 3.4), the right-coset incidence is verified independently. The conclusion $|Hg \cap V_{24}| = 4$ is established by the analogous explicit enumeration. The complementary count $|gH \setminus V_{24}| = |gH| - 4 = 16$ follows immediately, and similarly for right cosets. $\square$

Theorem 5.2 (τ -independence). The conclusion of Theorem 5.1 is independent of the choice of order-10 element τ used to construct the bulk subgroup H . Concretely, for each of the 24 order-10 elements $\tau \in G$, the resulting bulk subgroup $H_\tau := (K_\tau = 72) \cup (K_\tau = 0)$ is binary-dihedral of order 20, and every left and right coset of H_τ in G contains exactly 4 vertices of V_{24} . The 24 choices yield exactly 6 distinct subgroups H_τ , all of which are conjugate in G .

Proof. For each of the 24 order-10 elements, the verification script constructs H_τ , partitions G into the six H_τ -cosets, and checks the 4-fixed/16-mobile incidence with respect to V_{24} . The enumeration produces six distinct H_τ , all order-20 binary dihedral, all G -conjugate (by inspection of the orbit of H under conjugation by elements of G). The 4/16 incidence is uniform across all 24 choices. $\square$

Table 1: Coset incidence on $G = 2I$ with $H = \text{Dic}_5$ and V_{24} the σ -fixed sublattice. The incidence is uniform across left and right coset families; H is non-normal so this is not a quotient identity.

side	coset	$ C $	$ C \cap V_{24} $	$ C \setminus V_{24} $	S/A
left	0 ($= H$)	20	4	16	1/4
left	1	20	4	16	1/4
left	2	20	4	16	1/4
left	3	20	4	16	1/4
left	4	20	4	16	1/4
left	5	20	4	16	1/4
right	0 ($= H$)	20	4	16	1/4
right	1	20	4	16	1/4
right	2	20	4	16	1/4
right	3	20	4	16	1/4
right	4	20	4	16	1/4
right	5	20	4	16	1/4

6 The structural coefficient

Definition 6.1 (Anchor and channel counts). *For a subset $C \subseteq G$, set $S(C) := |C \cap V_{24}|$, $A(C) := |C \setminus V_{24}|$. We refer to $S(C)$ as the anchor count and $A(C)$ as the channel count of C .*

Theorem 6.1 (1/4 identity). *For every coset C of H (left or right) in G , $\frac{S(C)}{A(C)} = \frac{1}{4}$. For the union B of the five non-bulk left cosets, $S(B) = 20$, $A(B) = 80$, and $S(B)/A(B) = 1/4$.*

Proof. Combine Theorem 5.1 (numerator and denominator 4 and 16 per coset) with the partition of B into five disjoint cosets (S and A are additive over disjoint unions). $\square$

7 Scope

Scope.

- **Established (exact, unconditional):** the rational identity $|gH \cap V_{24}|/|gH \setminus V_{24}| = 1/4$ on the finite group $G = 2I$ with $H = \text{Dic}_5$, for both left and right cosets, with H self-normalizing and non-normal; and the τ -independence of the identity over all 24 order-10 choices.
- **Interpretive:** the identification of S with anchor information bits and A with horizon channels. This is offered as a definition with motivation, not as a theorem. A different interpretation may yield a different physical content from the same arithmetic identity.
- **Out of scope:** cascade-unit calibration to physical Planck units; the canonical τ_σ involution (assigned to a companion paper); derivation of the Hawking spectrum; Friedmann-equation analog; cosmological observables; ontology of V_{600} . The numerical match between the structural ratio $1/4$ and the Bekenstein–Hawking coefficient $1/4$ is observed without claim; the unit-calibration mapping required to lift this from a numerical coincidence to a quantitative physical derivation is not constructed here.

8 Comparison with existing microscopic derivations

We do not claim superiority over the established microscopic derivations; we report a methodologically distinct route. The following table summarises the relationship.

Approach	Mathematical content	Output (in natural units, $\hbar = 1$)
Strominger–Vafa [Strominger and Vafa, 1996]	D-brane partition function	$S/A = 1/4$
Ashtekar–Baez–Krasnov [Ashtekar et al., 2000]	spin-network state count	$S/A = 1/4$ (after standard L)
This paper	finite-group incidence on $2I$	$S/A = 1/4$ (dimensionless)

The incidence approach computes a dimensionless ratio. To match physical units, one would need to identify the cascade “area” unit (one σ -mobile vertex) with a Planck-area count and the cascade “entropy” unit (one σ -fixed vertex) with one bit of horizon information. The calibration constant is not derived here; it is the analogue of the $\hbar G c^{-3}$ scaling in the standard formula. What is established unconditionally is that the structural ratio is rational and equal to $1/4$.

9 Conclusion

The Bekenstein–Hawking coefficient $1/4$ has a finite-arithmetic analogue as a coset-incidence identity on the binary icosahedral group $2I$, namely $|gH \cap V_{24}|/|gH \setminus V_{24}| = 4/16 = 1/4$ for every coset of the binary dihedral subgroup $H = \text{Dic}_5$ against the σ -fixed 24-cell sublattice V_{24} , with both left and right coset families behaving identically despite the non-normality of H , and with the identity τ -independent across all 24 admissible bulk constructions. The result is exact ($\mathbb{Q}(\sqrt{5})$ arithmetic, no floating point), parameter-free, and verifiable by the accompanying script. We state explicitly that this is a structural identity rather than a derivation of physical black-hole thermodynamics, and we leave the cascade-unit calibration to subsequent work.

A Verification certificate

The companion script `verify.py` performs the following exact checks and exits with status 0 when all assertions hold:

1. $|V_{600}| = 120$.
2. Unit norm $|v|^2 = 1$ for all 120 vertices.
3. **Exhaustive** group closure: every product $v_i v_j$ across all $120^2 = 14,400$ pairs lies in V_{600} .
4. T_7 -cycle decomposition: 12 cycles of length 10; K-multiset $\{72:1, 0:1, 52:5, 20:5\}$.
5. $|H| = 20$, $1 \in H$, H closed under multiplication (all 400 $H \times H$ products), H self-normalizing in G .
6. Explicit Dic_5 generators (a, b) with $a \in H$ of order 10, $b \in H$ satisfying $b^2 = a^5$ and $bab^{-1} = a^{-1}$, and $\langle a, b \rangle = H$ verified by closure enumeration.
7. $|V_{24}| = 24$ and $V_{24} = V_{600} \cap \sigma(V_{600})$, with $|H \cap V_{24}| = 4$ and $|V_{24} \setminus H| = 20$.
8. Per-coset incidence table for both left and right cosets (Table 1), with explicit 4-element intersection indices printed for each coset.
9. Aggregate boundary $S/A = 20/80 = 1/4$.
10. τ -independence sweep over all 24 order-10 elements $\tau \in G$: each yields a binary-dihedral H_τ of order 20 with the explicit (a, b) presentation, partitions G into six left and six right cosets of size 20 each with uniform 4/16 incidence, and the 24 choices yield exactly 6 distinct H_τ subgroups all G -conjugate to the canonical H .

The script depends only on the Python standard library (`fractions`, `itertools`, `collections`) and on the shared `vfd_v600` package (`papers/v600-programme/lib`).

Reproducibility command (from the repository root):

```
PYTHONPATH=papers/v600-programme/lib \
python3 papers/bekenstein-incidence/verify.py
```

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